T.I.P. TECHNOLOGICAL INSTITUTE OF THE PHILIPPINES

COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010	**SECTION:**	CPE21S2
DATE PERFORMED:	07/17/26	**DATE SUBMITTED:**	07/17/26
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ACTIVITY NO. 1 **Hands-on Activity 1.1 Basic C++ Programming**

1. PROCEDURE OUTPUT

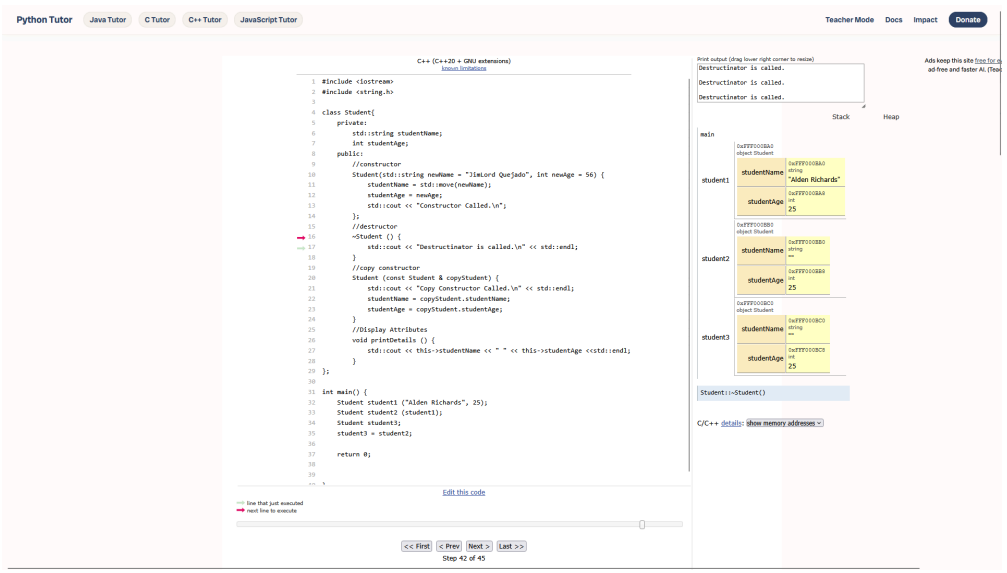
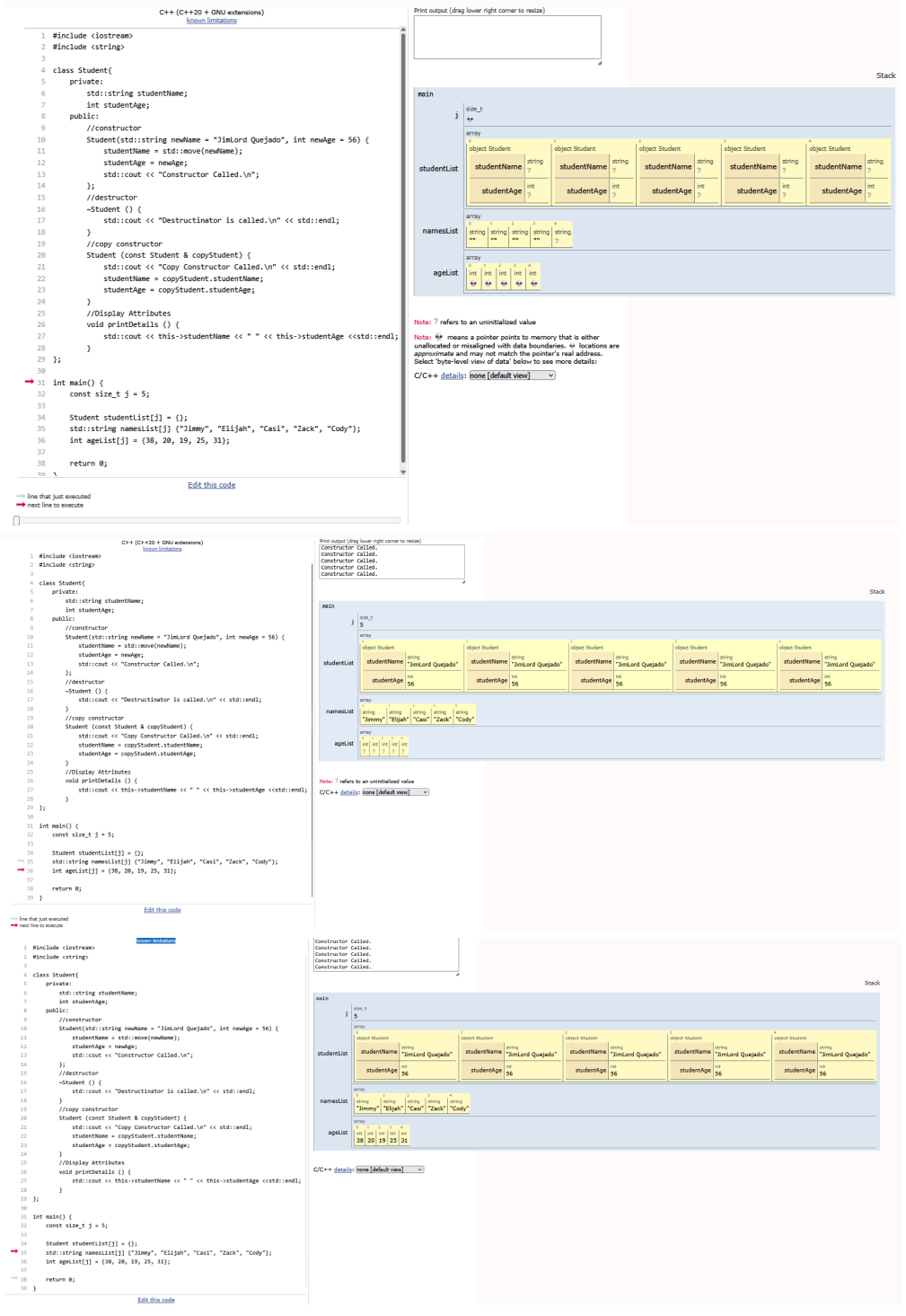
Screenshot	
Observation	Once code is running, classes and objects are initialized, The student Alden Richards with an age of 25 is created and moved to Student3 and again to Student 1. Afterwards, JimLord Quejado is created and is placed at the memory of Student3 and remains in place until the codes' runtime ends. Both of them are then destroyed at the end of the program.

Table 2-1. Initial Driver Program

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Screenshot	
Observation	During the runtime of the program, the constructor is called for each array which I expected because of the open brackets {}. Thus, resulting with each block being filled with the student

	JimLord Quejado and his age while being separated from the namesList and agelist that prints the names of the student from the array.

Table 2-2. Modified Driver Program with Student Lists

Loop A

main.cpp > main()

```cpp
#include <iostream>
#include <string>

class Student{
    private:
        std::string studentName;
        int studentAge;
    public:
        //constructor
        Student(std::string newName = "JimLord Quejado", int newAge = 56) {
            studentName = std::move(newName);
            studentAge = newAge;
            std::cout << "Constructor Called.\n";
        };
        //destructor
        ~Student () {
            std::cout << "Destructinator is called.\n" << std::endl;
        }
        //copy constructor
        Student (const Student & copyStudent) {
            std::cout << "Copy Constructor Called.\n" << std::endl;
            studentName = copyStudent.studentName;
            studentAge = copyStudent.studentAge;
        }
        //Display Attributes
        void printDetails () {
            std::cout << this->studentName << " " << this->studentAge <<std::endl;
        }
};

int main() {
    const size_t j = 5;

    Student studentList[j] = {};
    std::string namesList[j] = {"Jimmy", "Elijah", "Casi", "Zack", "Cody"};
    int ageList[j] = {38, 20, 19, 25, 31};

    return 0;
}
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

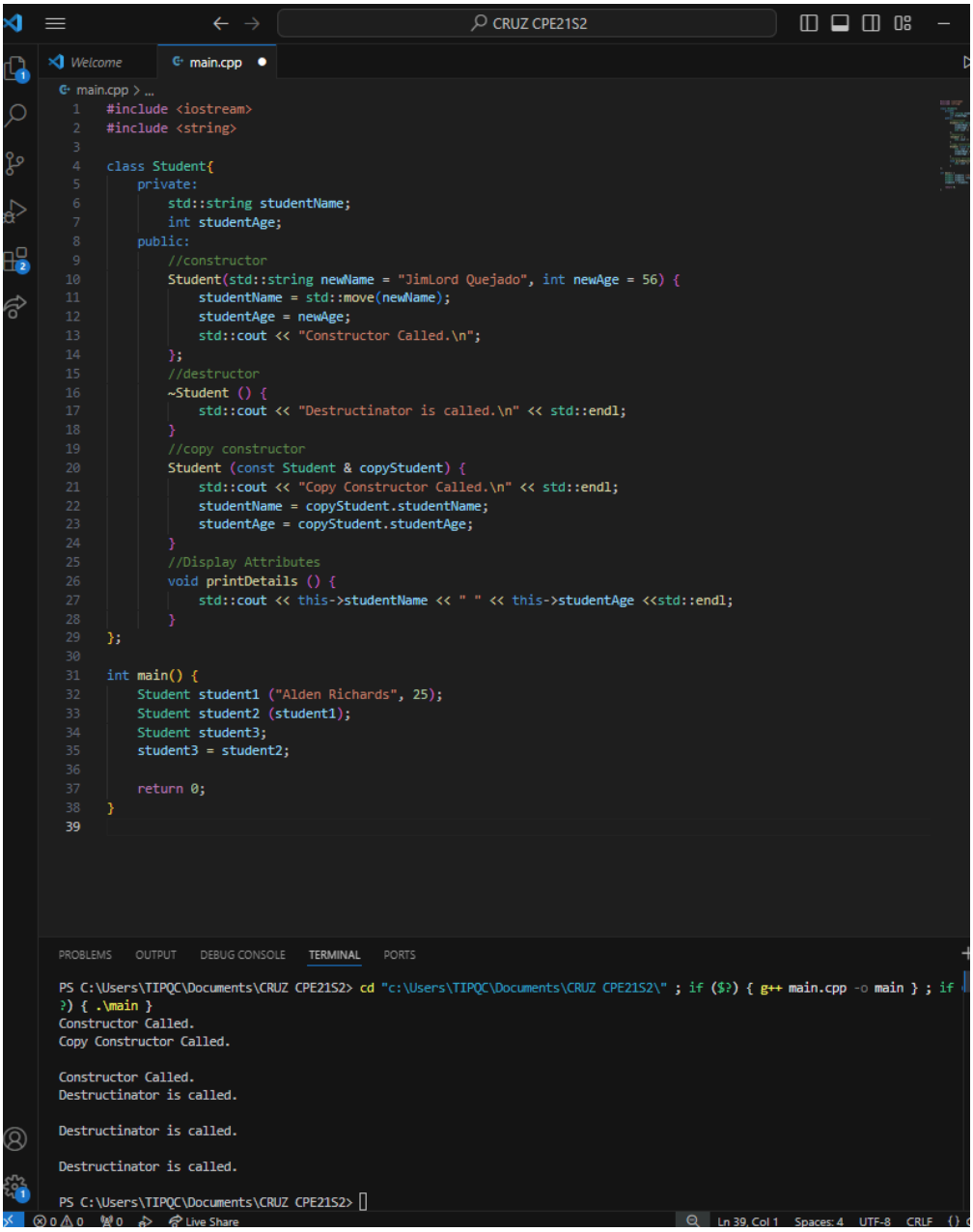
PS C:\Users\TIPQC\Documents\CRUZ CPE21S2> cd "c:\Users\TIPQC\Documents\CRUZ CPE21S2\" ; if ($?) { g++ main.cpp -o ma
?) { .\main }
Constructor Called.
Copy Constructor Called.

Constructor Called.
Destructinator is called.

Destructinator is called.

Destructinator is called.

PS C:\Users\TIPQC\Documents\CRUZ CPE21S2>
```

Observation	The program runs as intended and displays messages that the Constructor is called and Destructor is called. No errors were met and Python Tutor was used to visualize how the objects are created and moved around the program and memory.
Loop B	
Observation	Again, the program runs as intended and displays messages that the Constructor is called and Destructor is called. No errors were met and Python Tutor was used to visualize how the objects are created and moved around the program and memory.

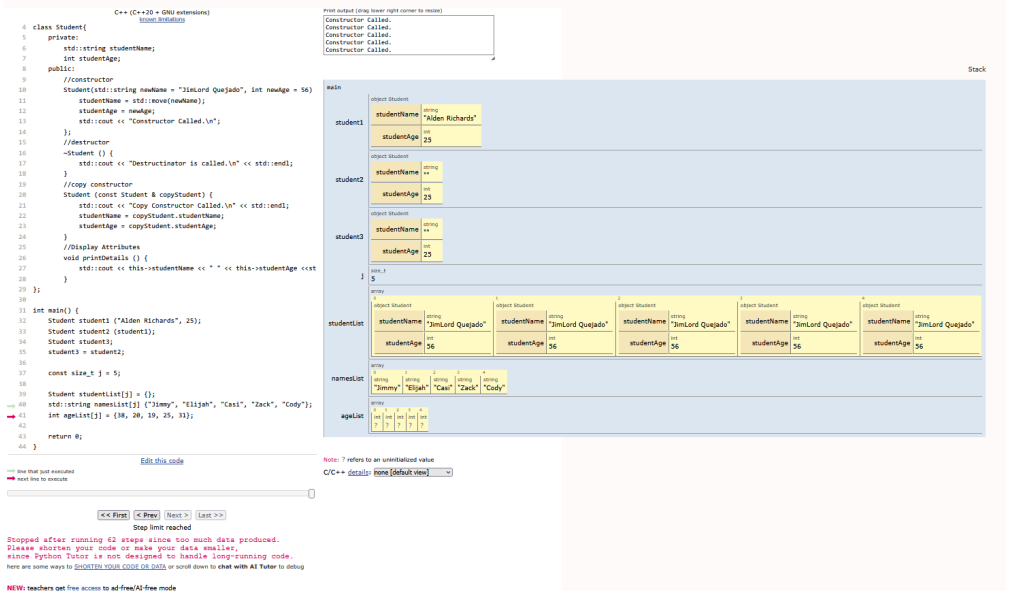
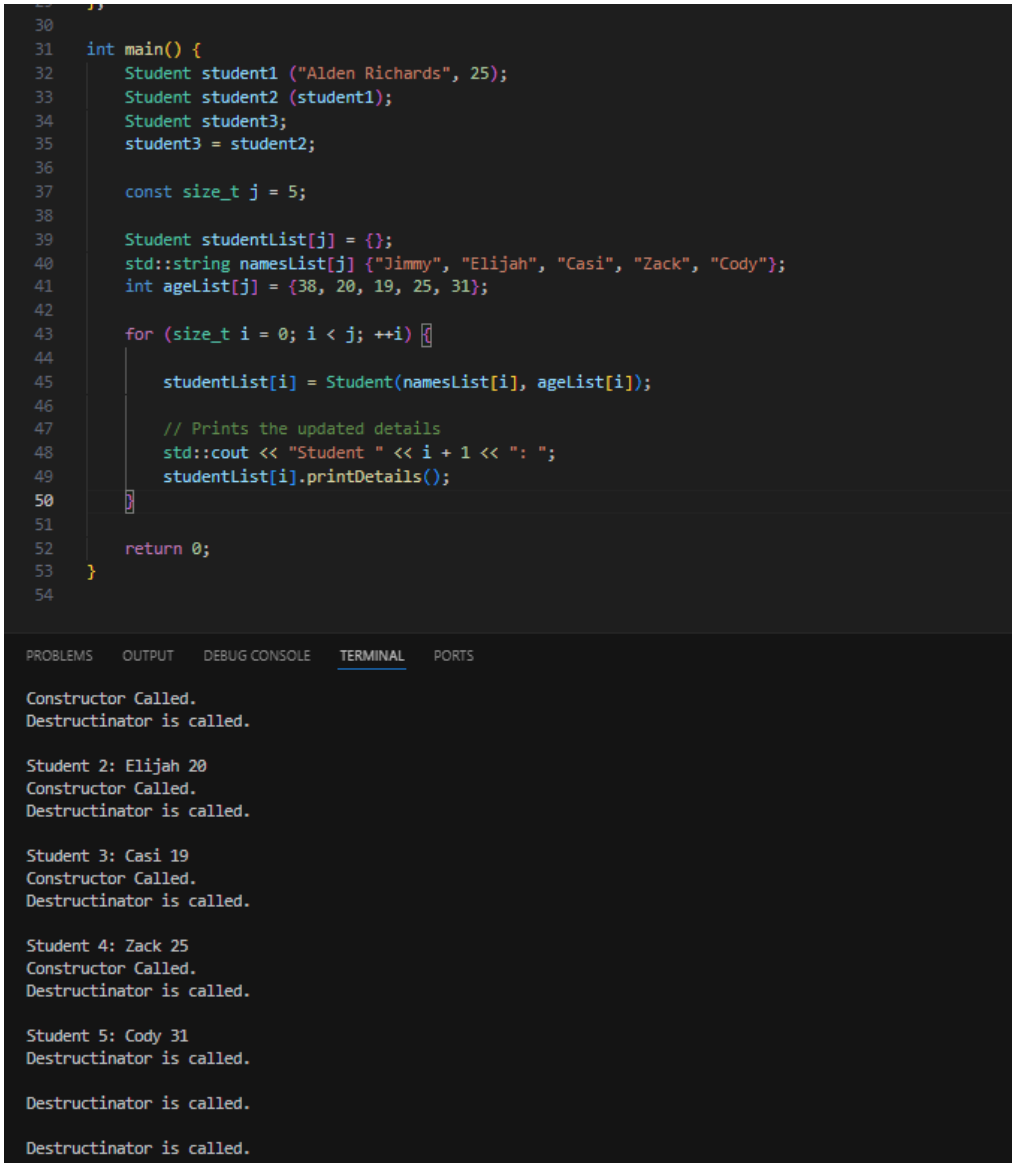
Output	
Observation	The program runs perfectly and no errors were met. Additionally, the code outputs multiple Constructor and Destructor is called. While the created objects are then deleted or removed from the memory at the end of the runtime of the program. All students names are then placed properly in their respective blocks as seen in the output with the student Jimlord Quejado noticeable filling the entire student list while student 2 and 3 remains empty.

Table 2-3. Final Driver Program

Modifications	
Observation	The modification applied is the For loop that outputs the name of the students.

```
30
31  int main() {
32      Student student1 ("Alden Richards", 25);
33      Student student2 (student1);
34      Student student3;
35      student3 = student2;
36
37      const size_t j = 5;
38
39      Student studentList[j] = {};
40      std::string namesList[j] {"Jimmy", "Elijah", "Casi", "Zack", "Cody"};
41      int ageList[j] = {38, 20, 19, 25, 31};
42
43      for (size_t i = 0; i < j; ++i) {
44
45          studentList[i] = Student(namesList[i], ageList[i]);
46
47          // Prints the updated details
48          std::cout << "Student " << i + 1 << ": ";
49          studentList[i].printDetails();
50      }
51
52      return 0;
53  }
54
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

Constructor Called.
Destructinator is called.

Student 2: Elijah 20
Constructor Called.
Destructinator is called.

Student 3: Casi 19
Constructor Called.
Destructinator is called.

Student 4: Zack 25
Constructor Called.
Destructinator is called.

Student 5: Cody 31
Destructinator is called.

Destructinator is called.

Destructinator is called.
```

Table 2-4. Modifications/Corrections Necessary

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

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I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Gemini

Purpose of Use:

Explanation of arrays, pointers, and code assistance.

Extent of Use:

Gemini was used to explain how function prototyping works and assistance in fixing errors within my code. I confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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